

D876

Cell Biology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A biology lab models two cube-shaped cells in gelatin dyed with food coloring. One has 2 μm sides and the other has 4 μm sides. The models took an afternoon to set in the refrigerator. What is the SA:V ratio of the larger cell?

- A. 6 : 1, since the ratio is the same for every cube
- B. 1.5 : 1
- C. 0.67 : 1
- D. 3 : 1

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

What side length of a cube-shaped cell gives an SA:V ratio of 2:1?

- A. 2 units
- B. 3 units
- C. 12 units
- D. 6 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Intestinal cells grown on a membrane insert from a supplier in Ohio take up a nutrient even when its concentration inside the cells is ten times higher than outside. The uptake uses a carrier protein and stops when a drug blocks ATP production. How is the nutrient transported?

- A. Simple diffusion
- B. Active transport
- C. Facilitated diffusion, because a carrier protein is involved
- D. Passive transport, since all membrane transport is passive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Red blood cells from a blood bank sample that expires next week are placed in 3% NaCl. Normal body fluid is about 0.9% NaCl. What happens to the cells?

- A. They swell as water enters.
- B. They stay the same size.
- C. They swell because salt rushes into the cells.
- D. They shrink as water leaves them.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which kind of solution makes an animal cell swell and possibly burst?

- A. Any salt solution
- B. Hypertonic
- C. Isotonic
- D. Hypotonic

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A cell line kept in a lab incubator set to 37 °C has a mutation that makes p53 nonfunctional. The cells are exposed to radiation that damages their DNA. The lab's incubator was replaced in the spring. What is the most likely result?

- A. Cells arrest at G1/S until the DNA is repaired.
- B. Apoptosis increases sharply.
- C. Cells continue dividing with damaged DNA.
- D. The spindle checkpoint stops the cells in metaphase.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Pancreatic cells in a culture dish with a pink-orange medium make insulin, which they release outside the cell. Researchers labeled the protein with a green fluorescent tag. What path does newly made insulin follow?

- A. Rough ER → Golgi → secretory vesicle → cell membrane
- B. Smooth ER → lysosome → cell membrane
- C. Golgi → rough ER → vesicle → cell membrane
- D. Free ribosome → cytosol

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A lab wants to see individual ribosomes, about 25 nm across, inside a cell. Its light microscope has a new 100× oil-immersion lens and a digital camera. Which tool is needed?

- A. A scanning electron microscope
- B. The light microscope at higher magnification
- C. A fluorescence light microscope
- D. A transmission electron microscope

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A structure is about 1 μm (1,000 nm) across. What is the simplest microscope that can resolve it?

- A. A light microscope
- B. A scanning electron microscope
- C. A transmission electron microscope, since anything cellular needs one
- D. The naked eye

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A technician mixes a fibroblast culture passaged three days ago 1:1 with trypan blue stain. She counts 200 cells total across 4 large hemocytometer squares (each 0.1 μL). The flask holds bright red medium. What is the cell concentration of the original culture?

- A. 2.5×10^5 cells/mL
- B. 4.0×10^6 cells/mL
- C. 1.0×10^6 cells/mL
- D. 5.0×10^5 cells/mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A culture holds 1.0×10^6 cells/mL. What volume should be added to a well to seed 2.0×10^5 cells?

- A. 0.20 mL
- B. 0.020 mL
- C. 5.0 mL
- D. 2.0 mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A lab ordered a restriction enzyme in 5,000-unit vials and stores it at -20°C . It plans to insert a gene into a plasmid and later check the sequence with an online database search. What step makes the gene and plasmid able to join?

- A.** Use DNA polymerase to glue the pieces.
- B.** Heat-shock the bacteria, which joins the DNA pieces.
- C.** Cut both with the same restriction enzyme, then seal the matching sticky ends with DNA ligase.
- D.** Cut the gene and the plasmid with two different enzymes that leave unrelated ends.

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A biology lab models two cube-shaped cells in ~~gelatin dyed with food coloring~~. One has 2 μm sides and the other has 4 μm sides. ~~The models took an afternoon to set in the refrigerator.~~ What is the SA:V ratio of the larger cell?

Target word: SA:V of the 4 μm cube **Decides:** $SA = 6 \times 4^2 = 96$; $V = 4^3 = 64$.

A	Absolute language	SA:V = 6/side, so it changes with size.
B	Correct	$96 / 64 = 1.5$.
C	Inverted operation	$64 / 96$ gives V:SA.
D	Wrong stem number	That is the 2 μm cube ($24 / 8$).

2. Answer B Concept 1 • transfer

What side length of a cube-shaped cell gives an SA:V ratio of 2:1?

Target word: solve for side length **Decides:** $SA:V = 6/s$.

A	Wrong stem number	A 2-unit cube has $SA:V = 3:1$.
B	Correct	$6 / 3 = 2$ (SA 54, V 27).
C	Inverted operation	Multiplies 6×2 instead of dividing.
D	Dropped element	A 6-unit cube has $SA:V = 1:1$.

3. Answer B Concept 2

Intestinal cells grown on a membrane insert ~~from a supplier in Ohio~~ take up a nutrient even when its concentration inside the cells is ten times higher than outside. The uptake uses a carrier protein and stops when a drug blocks ATP production. How is the nutrient transported?

Target word: transport type **Decides:** Movement against the gradient that depends on ATP.

A	Adjacent category	Moves only down the gradient, without proteins.
B	Correct	Moving against a gradient requires energy.
C	Confusion set	Facilitated diffusion uses carriers but only moves down the gradient.
D	Absolute language	Active transport is a major class of membrane transport.

4. Answer D Concept 3

Red blood cells ~~from a blood bank sample that expires next week~~ are placed in 3% NaCl. Normal body fluid is about 0.9% NaCl. What happens to the cells?

Target word: cell response to a hypertonic solution **Decides:** 3% is hypertonic to the cell.

A	Inverted operation	That happens in a hypotonic solution.
B	Confusion set	That happens only in an isotonic ($\approx 0.9\%$) solution.
C	Adjacent category	In osmosis, water moves, not the salt.
D	Correct	Water moves out toward the higher solute concentration.

5. Answer D Concept 3 · transfer

Which kind of solution makes an animal cell swell and possibly burst?

Target word: name the solution from the result **Decides:** Water enters when outside solute is lower.

A	Absolute language	Only a salt solution weaker than the cell's interior causes swelling.
B	Inverted operation	Causes shrinking.
C	Confusion set	No net water movement.
D	Correct	Lower outside solute draws water in.

6. Answer C Concept 4

A cell line ~~kept in a lab incubator set to 37 °C~~ has a mutation that makes p53 nonfunctional. The cells are exposed to radiation that damages their DNA. ~~The lab's incubator was replaced in the spring.~~ What is the most likely result?

Target word: effect of losing p53 after DNA damage **Decides:** Without p53, the G1/S checkpoint does not stop damaged cells.

A	Inverted operation	That is what working p53 does.
B	Adjacent category	p53 is a key trigger for apoptosis, so apoptosis decreases.
C	Correct	The checkpoint that would halt or kill them is missing.
D	Confusion set	The spindle checkpoint monitors chromosome attachment, not DNA damage.

7. Answer A Concept 5

Pancreatic cells in a culture dish ~~with a pink-orange medium~~ make insulin, which they release outside the cell. ~~Researchers labeled the protein with a green fluorescent tag.~~ What path does newly made insulin follow?

Target word: route of a secreted protein **Decides:** Secreted proteins use the endomembrane pathway.

A	Correct	The standard secretory pathway.
B	Confusion set	Smooth ER makes lipids; lysosomes digest material.
C	Inverted operation	Reverses the ER and Golgi.
D	Adjacent category	Free ribosomes make proteins that stay in the cytosol.

8. Answer D Concept 6

A lab wants to see individual ribosomes, about 25 nm across, inside a cell. Its light microscope ~~has a new 100× oil immersion lens and a digital camera.~~ Which tool is needed?

Target word: microscope for a 25 nm interior structure **Decides:** 25 nm is far below the ~200 nm light limit.

A	Confusion set	SEM images surfaces, not structures inside the cell.
B	Dropped element	More magnification does not beat the resolution limit.
C	Adjacent category	Standard fluorescence is still limited to about 200 nm.
D	Correct	TEM resolves nanometer-scale internal structures.

9. Answer A Concept 6 • transfer

A structure is about 1 μm (1,000 nm) across. What is the simplest microscope that can resolve it?

Target word: lowest-tech tool that works **Decides:** 1,000 nm is above the ~200 nm light limit.

A	Correct	Light microscopy resolves down to about 200 nm.
B	Adjacent category	Works, but is not the simplest option.
C	Absolute language	Many cell structures are large enough for light.
D	Dropped element	The unaided eye resolves only about 0.1 mm.

10. Answer C Concept 7

A technician mixes a fibroblast culture ~~passed three days ago~~ 1:1 with trypan blue stain. She counts 200 cells total across 4 large hemocytometer squares (each 0.1 μL). ~~The flask holds bright red medium.~~ What is the cell concentration of the original culture?

Target word: cells per mL **Decides:** Average 50 per square $\times 10^4 \times$ dilution 2.

A	Inverted operation	Divides by the dilution factor instead of multiplying.
B	Wrong stem number	Uses the total 200 as if it were the per-square average.
C	Correct	$(200 / 4) \times 10^4 \times 2 = 1.0 \times 10^6$.
D	Dropped element	Leaves out the 1:1 dilution factor.

11. Answer A Concept 7 • transfer

A culture holds 1.0×10^6 cells/mL. What volume should be added to a well to seed 2.0×10^5 cells?

Target word: solve for volume **Decides:** Volume = cells needed / concentration.

A	Correct	$2.0 \times 10^5 / 1.0 \times 10^6 = 0.20 \text{ mL}$.
B	Confusion set	Would seed only 2.0×10^4 cells.
C	Inverted operation	Divides concentration by cells needed.
D	Wrong stem number	Off by a factor of ten; would add 2.0×10^6 cells.

12. Answer C Concept 8

A lab ordered a restriction enzyme ~~in 5,000-unit vials and stores it at -20°C~~ . It plans to insert a gene into a plasmid ~~and later check the sequence with an online database search~~. What step makes the gene and plasmid able to join?

Target word: how insert and vector join **Decides:** Matching sticky ends from the same enzyme, sealed by ligase.

A	Confusion set	Polymerase copies DNA; ligase joins fragments.
B	Adjacent category	Heat shock moves the finished plasmid into cells; it does not join DNA.
C	Correct	Complementary ends pair; ligase forms the backbone bonds.
D	Inverted operation	Unmatched ends will not pair.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Surface-area-to-volume ratio	As a cell grows, volume rises faster than surface area, so SA:V falls; for a cube, SA:V = 6/side.	Thinking bigger cells have more surface per unit volume.	Cube-shaped cells of different sizes.
2	Active vs. passive transport	Moving a substance against its concentration gradient requires energy (ATP or a coupled gradient); diffusion and facilitated diffusion only move down the gradient.	Calling any carrier-protein transport active.	Uptake that stops when ATP production is blocked.
3	Tonicity and osmosis	Water moves toward higher solute concentration: animal cells shrink in hypertonic solutions and swell (may burst) in hypotonic ones.	Thinking water follows toward lower solute, or that salt moves instead of water.	A cell placed in a solution of stated concentration.
4	Cell-cycle checkpoints and p53	Checkpoints halt the cycle when DNA is damaged; p53 stops the cycle at G1/S or triggers apoptosis, and its loss lets damaged cells keep dividing.	Thinking a damaged cell will stop dividing whether or not p53 works.	A mutated tumor-suppressor gene plus DNA damage.
5	Secretory pathway	Secreted proteins are made on rough ER ribosomes, modified in the Golgi, and shipped in vesicles for exocytosis.	Reversing ER and Golgi, or placing synthesis on free ribosomes.	A protein that leaves the cell, such as a hormone.
6	Resolution sets the microscope	Light microscopes resolve down to about 200 nm; smaller structures need electron microscopy (TEM for interiors, SEM for surfaces). Magnification alone does not add detail.	Thinking a higher-power objective will reveal anything.	The size of the structure to be imaged.
7	Hemocytometer counts	Cells/mL = average count per large square $\times 10^4 \times$ dilution factor.	Forgetting the dilution factor or using the total count as the average.	A count across several squares after mixing with a stain.
8	Recombinant DNA with restriction enzymes	Cut the vector and the gene with the same restriction enzyme so sticky ends match, then seal with DNA ligase.	Using different enzymes, or thinking polymerase joins the pieces.	Inserting a gene into a plasmid.

Course scope (public description): Cell Biology examines the fundamental principles and techniques that underpin the structure, function, and growth of cells, encompassing cellular components, the cell cycle, and genetic information flow. The course emphasizes the importance of cellular processes and techniques such as microscopy, cell culture, genetic engineering, and bioinformatics, highlighting their relevance in understanding cellular biology's role in health and disease. This course is designed to provide students with a comprehensive understanding of how cells maintain homeostasis, metabolize, and perform specialized functions.